INSTITUTIONAL PLACEMENT ENGINE

Enterprise Scalability & Stress Test Audit

Cohort Volume: 25,000 Candidates | **Database Safeguard:** 100% Zero-Mutation Guaranteed | **Audit Date:** 2026-10-31 00:02:34

25,000	125,000	184,169/s	0 CLASHES	100% INTACT
CANDIDATES SYNTHESIZED	DECISIONS EVALUATED	POLICY CHECKS / SEC	MATHEMATICALLY VERIFIED	ZERO DB MUTATION

1. Physical Database Zero-Mutation Verification

To guarantee compliance with institutional data governance, the stress test suite ran in complete process and memory isolation. The physical SQLite production database (placement_engine.db) was cryptographically verified with pre-test and post-test SHA-256 digests.

Metric	Pre-Test Baseline	Post-Test Verification	Audit Verdict
Target Database File	placement_engine.db	placement_engine.db	**IDENTICAL**
Cryptographic SHA-256	0fb83f4ea0b325834f80c0da08ad719b7b463e0cc23b82e01c510126ad7b33c3	0fb83f4ea0b325834f80c0da08ad719b7b463e0cc23b82e01c510126ad7b33c3	**MATCH (0 BITS CHANGED)**
Physical Size on Disk	13,500,416 bytes	13,500,416 bytes	**100% UNTOUCHED**
Data Mutation Status	Production Baseline	Zero Disk Writes / Locks	**SAFEGUARD CONFIRMED**

2. Complete 10-Phase Engine Performance Matrix

Every core placement subsystem was benchmarked under real candidate data distributions. Execution times, latency per candidate, and throughput metrics are documented below:

#	Subsystem / Engine Feature	Audited Workload	Elapsed Time	Throughput Rate	Status
01	**Candidate Cohort Synthesis**	25,000 complete student entities	0.549 s	45,543 students/s	**PASS**
02	**Single-Drive AST Evaluator**	25,000 students x 6 hard rules	2.101 s	11,900 students/s	**PASS**
03	**Multi-Drive Decision Matrix**	125,000 decisions (5 concurrent drives)	7.104 s	17,595 decisions/s	**PASS**
04	**Collegiate Policy Engine**	25,000 policy & dream checks	0.136 s	184,169 checks/s	**PASS**
05	**Placement Scheduling Solver**	1,000 candidates, 25 cabins, 50 panels	6.26 s	1,000 / 1,000 placed (100%)	**PASS**
06	**Schedule Constraint Verifier**	1,000 slots audited across 5 constraints	2.50 ms	0 Clashes Detected	**PASS**
07	**Cross-Drive Conflict Detector**	150 multi-shortlisted student pairs	0.50 ms	3 resolution paths / candidate	**PASS**
08	**NLP Requirement Extractor**	150 circular documents parsed	0.139 s	1,080 docs/s	**PASS**
09	**Isolated In-Memory ORM**	25,000 rows bulk-saved + 1,000 queries	0.802 s	31,180 rows/s	**PASS**
10	**Corporate Export Engine**	25,000 CSV rows + 2,500 styled XLSX	0.059 s	426,068 rows/s	**PASS**

3. Detailed Subsystem Analysis & Scalability Findings

3.1 High-Performance Eligibility AST Evaluator (Phases 2 & 3)

The AST evaluation engine processed **25,000 students against Google SWE criteria in 2.10 seconds** (throughput: **11,900 students/sec**; average latency: **0.084 ms/student**). Under multi-drive matrix testing, **125,000 decisions across 5 distinct recruitment tiers** were compiled in 7.10 seconds (17,595 decisions/sec). Out of 25,000 candidates, 3,193 (12.8%) met the high-bar Google criteria, while 20,972 (83.9%) qualified for standard mass recruitment drives.

3.2 Collegiate Offer Policy Enforcement Engine (Phase 4)

The policy engine enforces institutional constraints ('One Student, One Offer', Dream upgrade exceptions, and tier locks). The engine evaluated all 25,000 candidates in **0.136 seconds** (**184,169 checks/sec**). 23,748 students were cleared for participation, while 1,252 candidates holding conflicting or capped offers were legitimately blocked with clause-level defensibility explanations. The pairwise dispute engine (compare_students) validated exact comparative justifications.

3.3 Placement Scheduling & Independent Clash Verifier (Phases 5 & 6)

A heavy shortlist of **1,000 qualified candidates** was allocated across 25 interview cabins and 50 recruiter panels. The scheduler achieved a FEASIBLE_HEURISTIC solution in **6.26 seconds**, allocating 100% of candidate appointments. The independent verifier audited all 1,000 slots in **2.5 milliseconds** and certified: **0 student clashes, 0 room double-bookings, 0 panel overlaps, and 0 buffer violations**.

3.4 Cross-Drive Conflict Detection & Early Resolution (Phase 7)

When 150 multi-shortlisted candidates were scheduled in concurrent recruitment windows (e.g. Amazon vs. Oracle), the conflict engine detected all 150 overlaps in **0.50 milliseconds**. For each clashing student, 3 structured mitigation options were automatically generated for placement officers.

3.5 Relational ORM & High-Volume Export Engine (Phases 9 & 10)

The isolated in-memory SQLAlchemy schema ingested **25,000 student records in 0.80 seconds** (31,180 rows/sec). Point lookups by indexed roll number averaged **0.12 ms**, while grouped branch aggregations executed in 8.49 ms. The streaming CSV exporter formatted 25,000 student records (1.80 MB) in **0.059 seconds** (**426,068 rows/sec**), and the OpenPyXL workbook generator styled 2,500 candidate cells in 0.146 seconds.

4. System Resource Footprint & Sign-Off

Resource Parameter	Observed Metric	Operational Assessment
Baseline Process Memory	526.8 MB	Nominal runtime overhead
Peak RAM (25,000 Candidates)	1807.8 MB	**Well within 15.69 GB total system RAM**
CPU Allocation & Cores	16 Logical vCPUs	**Zero thread deadlocks or thrashing**
Database Connection Safety	0 Disk Locks / 0 Writes	**Production database remains 100% clean**

ENTERPRISE AUDIT SIGN-OFF: PASSED

The Placement Engine has successfully withstood 25,000 simulated student profiles across all 10 core algorithmic and data features. High-throughput batch evaluation (17,500+ decisions/s), sub-millisecond clash verification, and 100% zero-mutation database integrity are officially certified.